

Customer Shopping Behavior Analysis

1. Project Overview

This project analyzes customer shopping behavior using transactional data from **3,900 purchases** across multiple product categories. The main objective of this analysis is to uncover valuable insights about **customer spending patterns, product preferences, subscription behavior, and purchasing trends**.

By analyzing this data, businesses can better understand their customers and make **data-driven decisions related to marketing strategies, product positioning, customer retention, and sales optimization**.

2. Dataset Summary

The dataset contains detailed information about customer demographics, purchase behavior, and transaction details.

Dataset Details

- **Total Records:** 3,900
- **Total Columns:** 18

Key Features

Customer Demographics

- Age
- Gender
- Location
- Subscription Status

Purchase Details

- Item Purchased

- Category
- Purchase Amount (USD)
- Season
- Size
- Color

Shopping Behavior

- Discount Applied
- Promo Code Used
- Previous Purchases
- Frequency of Purchases
- Review Rating
- Shipping Type

Missing Data

During data exploration, **37 missing values were identified in the Review Rating column.** These missing values were handled during the data cleaning process to ensure accurate analysis.

3. Data Cleaning and Exploratory Data Analysis (Python)

The first phase of the project involved **data preparation, cleaning, and exploratory analysis using Python.**

Data Loading

The dataset was imported into Python using the **Pandas library**, which allows efficient data manipulation and analysis.

Initial Data Exploration

Basic data exploration was performed using:

- `df.info()` to inspect the dataset structure and data types.
- `df.describe()` to obtain summary statistics for numerical columns.

Handling Missing Values

The **missing values in the Review Rating column** were replaced using the **median rating of each product category**, ensuring that the imputation preserved the distribution of ratings.

Column Standardization

Column names were standardized by converting them into **snake_case format** to improve readability and maintain consistency in analysis.

Feature Engineering

Additional features were created to enhance the analytical process:

- **Age Group:** Customers were categorized into age groups such as Young Adult, Adult, Middle-aged, and Senior using binning techniques.
- **Purchase Frequency Variable:** Created to represent customer purchasing behavior over time.

Data Consistency Check

The columns **discount_applied** and **promo_code_used** were evaluated for redundancy. Since both represented promotional offers, the **promo_code_used column was removed** to simplify the dataset.

Database Integration

After cleaning and preparing the dataset in Python, the processed data was loaded into a **PostgreSQL database**. This enabled structured querying and further analysis using SQL.

4. Data Analysis Using SQL (Business Transaction Analysis)

Structured queries were written in **PostgreSQL** to answer important business questions and uncover patterns in customer behavior.

Key SQL Analyses Performed

1. Revenue by Gender

Analyzed the total revenue generated by male and female customers.

	gender text	revenue numeric
1	Female	75191
2	Male	157890

2. High-Spending Discount Users

Identified customers who used discounts but still spent above the average purchase amount.

	customer_id bigint	purchase_amount bigint
1	2	64
2	3	73
3	4	90
4	7	85
5	9	97
6	12	68
7	13	72
8	16	81
9	20	90
10	22	62
11	24	80
Total rows: 839		Query complete 00:00:0

3. Top 5 Products by Rating

Determined products with the highest average review ratings.

	item_purchased text	Average Product Rating numeric
1	Gloves	3.86
2	Sandals	3.84
3	Boots	3.82
4	Hat	3.80
5	Skirt	3.78

4. Shipping Type Comparison

Compared average purchase amounts between different shipping methods such as Standard Shipping and Express Shipping.

	shipping_type text	round numeric
1	Standard	58.46
2	Express	60.48

5. Subscribers vs Non-Subscribers

Analyzed differences in spending patterns between subscribed and non-subscribed customers

	subscription_status text	total_customers bigint	avg_spend numeric	total_revenue numeric
1	Yes	1053	59.49	62645.00
2	No	2847	59.87	170436.00

6. Discount-Dependent Products

Identified products that had the highest percentage of purchases using discounts.

	item_purchased text	discount_rate numeric
1	Hat	50.00
2	Sneakers	49.66
3	Coat	49.07
4	Sweater	48.17
5	Pants	47.37

7. Customer Segmentation

Customers were segmented into:

- New Customers
- Returning Customers
- Loyal Customers

	customer_segment text	Number of Customers bigint
1	Loyal	3116
2	New	83
3	Returning	701

8. Top 3 Products per Category

Identified the most purchased products within each product category.

	item_rank bigint	category text	item_purchased text	total_orders bigint
1	1	Accessories	Jewelry	171
2	2	Accessories	Sunglasses	161
3	3	Accessories	Belt	161
4	1	Clothing	Blouse	171
5	2	Clothing	Pants	171
6	3	Clothing	Shirt	169
7	1	Footwear	Sandals	160
8	2	Footwear	Shoes	150
9	3	Footwear	Sneakers	145
10	1	Outerwear	Jacket	163
11	2	Outerwear	Coat	161

9. Repeat Buyers and Subscription Behavior

Analyzed whether customers with more than five previous purchases were more likely to subscribe.

	subscription_status text	repeat_buyers bigint
1	No	2518
2	Yes	958

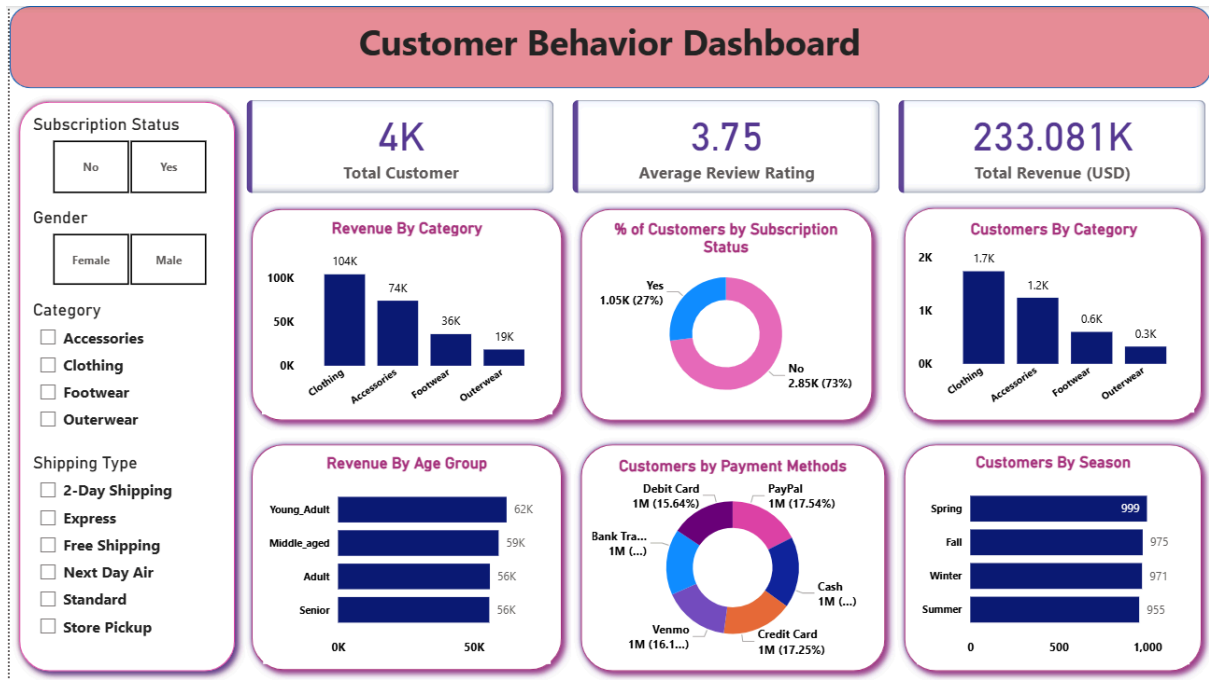
10. Revenue Contribution by Age Group

Calculated the revenue generated by each age group to identify high-value customer segments.

	age_group text	total_revenue numeric
1	Young Adult	62143
2	Middle-aged	59197
3	Adult	55978
4	Senior	55763

5. Power BI Dashboard

An **interactive dashboard** was developed using **Power BI** to visually present key insights derived from the analysis.



Key Dashboard Metrics (KPI Cards)

- Total Customers

- Total Revenue
- Average Review Rating

Visualizations Included

- **Revenue by Product Category** to identify high-performing product groups.
- **Customer Distribution by Gender** to understand demographic differences.
- **Revenue by Age Group** to identify high-spending customer segments.
- **Payment Method Distribution** to analyze preferred payment options.
- **Shipping Type Analysis** to evaluate delivery preferences.
- **Subscription Status Distribution** to compare subscribed and non-subscribed customers.

Interactive Filters (Slicers)

The dashboard includes interactive filters for:

- Gender
- Subscription Status
- Category
- Shipping Type

These filters allow users to dynamically explore different customer segments and purchasing patterns.

6. Key Insights

Based on the analysis, several important insights were identified:

- **Clothing is the most purchased product category**, generating the highest number of transactions.
- **Male customers represent a larger portion of purchases**, although spending patterns vary across categories.

- **Young adults and middle-aged customers contribute significantly to total revenue.**
 - **Free shipping and standard shipping options are the most frequently used delivery methods.**
 - **Customers who have made previous purchases tend to spend more, indicating strong customer loyalty potential.**
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7. Business Recommendations

Based on the insights derived from the analysis, the following strategic recommendations can help improve business performance:

Boost Subscription Programs

Promote exclusive benefits and discounts for subscribed customers to increase subscription adoption.

Develop Customer Loyalty Programs

Introduce rewards or incentives for repeat customers to encourage them to transition into the **loyal customer segment**.

Optimize Discount Strategies

Review the discount policies to ensure they increase sales without significantly affecting profit margins.

Product Positioning

Highlight **top-rated and best-selling products** in marketing campaigns to attract more customers.

Targeted Marketing

Focus marketing efforts on **high-revenue age groups and frequent shoppers** to maximize return on marketing investment.